

Suchman (2011). Practice and its overflows: Reflections on order and mess. *Tecnoscienza*, 2(1), 21-30.

If useful, my reading annotations may be found here:  [Suchman_2011_annotated.pdf](#)

Abstract

The issue of researchers as subjects that inherently shape the objects of their study through their interaction has been discussed frequently in science and technology studies (STS), especially in terms of problematizing the limits of the Cartesian subject/object model. In demonstrating how airline operations workers are situated in a larger system organized by the regulations and norms prescribed by the airline timetable schedule, it is apparent that the workers' and schedules' efforts are interrelated, with a subject/object model being insufficient in accounting for their relation. Likewise, researchers must understand themselves as an organizing or ordering force, like the airline schedule, that shapes the terms of research production in an interactive way, which has implications for our role in determining what is ultimately included/excluded in produced research.

Synthetic Notes

Article Goals

To interrogate the conceptual model of viewing systems in terms of an order/mess dichotomy in favor of viewing systems, both in terms of those being studied and the system of research itself, as an assemblage with constituent parts that, through their relations, work to define and fulfill particular goals.

Methods for Achieving the Article's Goals

Suchman draws on a case study exploring how airline timetable schedules act as an organizing point for defining how airport operations should be conducted. Through an ethnomethodological approach including the observation of operations workers as they situate themselves in interaction with the schedule as it is presented via a computer system, Suchman illustrates how the seemingly ordered expression of the airline timetable actually requires the

workers to anticipate and negotiate the messiness of forces that might prevent adherence to the prescribed norms outlined by the schedule.

Central Argument

Just as the operations workers reveal the messiness that necessitates ongoing action from all actors/actants within a system/network/assemblage, (STS) researchers must recognize themselves as engaging in an ordering *practice* that shapes what is included or excluded from that practice.

Disciplinary Contributions

This article is directed toward those in STS, or science and technology studies. Suchman's larger body of scholarship touches on a variety of disciplinary intersections, including STS, anthropology, and feminist studies. Her work would be beneficial to anyone interested in technical communication, human-computer interaction, embodiment as it relates to machines and technical systems, and ethnographic or ethnomethodological research within technical spaces.

All the same, Suchman offers an important lesson in this article that is relevant to anyone doing research work, as she explores the role of the researcher as interpreter in all aspects of the research production process, including (though certainly not limited to) their selection of theoretical frameworks, methodological approaches, implementation of those methodologies in their analysis, interpretation of their results, and presentation of those findings within their disciplinary communities.

Key Terms & Concepts

Apparatus

In the case of Karen Barad's¹ work, the notion of the apparatus is extended to include considerations of the body and embodied practice in a way that expands beyond human bodies. Marshall and Alberti² offer an explanation that I find very useful:

¹ Barad, K. (2007). *Meeting the Universe halfway: Quantum physics and the entanglement of matter and meaning*. Duke University Press.

² Marshall, Y., and Alberti, B. (2014). A matter of difference: Karen Barad, ontology, and archaeological bodies. *Cambridge Archaeological Journal*, 24(1), 19-36.

“Barad’s concepts of ‘phenomenon’ and ‘apparatus of bodily production’ reformulate all bodies, not just the biological, as complex entanglements whose properties or attributes are products or effects of phenomena rather than pre-existing those phenomena” (p. 22).

Centers of Coordination

These are the central organizing sites from which a system is primarily ordered and maintained. In her offering of this concept, Suchman draws on Latour’s “center of calculation”³, or the sites in which knowledge is produced through the accumulation and circulatory movement of information.

Disciplinary Technology

As Suchman alludes to, the notion of a disciplinary technology draws on Foucault’s⁴ theorizing on issues of power, regulation, and punishment. The Cambridge Foucault Lexicon⁵ outlines his views on disciplinary technologies:

“At the core of the disciplinary process are three distinct technologies – hierarchical observation, normalizing judgment, and examination – by which individuals are trained and transformed into docile and useful bodies. The process is grounded on the norm, which functions as a principle of coercion and correct action, a standard characterized by basic aspects of ‘time, activity, behavior, speech, body, attitude, and sexuality’...” (p. 113).

Indexicality

In linguistics, indexicality⁶ refers to the process by which a linguistic choice is imbued with social, stylistic, or pragmatic meaning. Here, there is a difference between things that are indexed *directly* or *indirectly*. For example, features associated with Southern language varieties often index multiple social meanings – the use of iconic lexical items (i.e. “y’all”), phonetic features (i.e. /ai/ ungliding or monophthongization), or grammatical features (i.e. double modals like “might could”) are often associated with views of Southerners as backwards

³ Latour, B. (1987). *Science in action*. Harvard University Press.

⁴ Foucault, M. (1979). *Discipline and punish: The birth of the prison*. Vintage.

⁵ Lawy, L., & Nale, J. (Eds.) (2015). *The Cambridge Foucault lexicon*. Cambridge University Press.

⁶ Ochs, E. (1992). Indexing gender. In A. Duranti & C. Goodwin (Eds.), *Rethinking context: Language as an interactive phenomenon* (pp. 335-358). Cambridge University Press.

thinking, conservative, uneducated, and improper. Of course, it is not that “y’all” actually carries any direct meaning that a person is unintelligent, but because Southerners have become socially associated over time with the use of “y’all” and a lack of education, those two things become linked together. In other words, “y’all” comes to *indirectly* index a lack of education.

Semiotics

Also in the field of linguistics, the study of semiotics is focused on links between *signs*, or linguistic representations, and their *signifiers*, or the meaning that is signalled by the sign that can be interpreted. There is some similarity between indexical theories of language and semiotics, but semiotics is somewhat more broad in that it allows for analysis of linguistic features being linked to a broader range of meanings, both directly and indirectly, that come to be associated with a given sign.

Discussion Questions

1. What does Suchman mean when she says that the airline timetable schedule has an “indexical relationship to the order of events that it prescribes... work that it presupposes, but does not itself fully specify” (p. 25)? Where might we find a distinction between the direct indexical meaning that the schedule conveys and the additional, unspoken, or indirect indexical meaning that it carries?
2. Suchman focuses her discussion of the relational nature of the airline schedule network through discussion of the *apparatus* and the *center of coordination*. In what ways does her use of these conceptual frameworks align with or potentially differ from those offered in the other readings we have looked at. Would her findings change if we view this set of relations with a more fine-grained focus on their collection as an assemblage or from the lens of actor-network theory? How so?

Further Readings

Goodwin, C. (1994). Professional vision. *American Anthropologist*, 96(3), 606-633.

This chapter explores the establishment of professional discourses and analytical lenses and their implications for directing the embodiment and behaviors or practices of those

working in various settings, including through the analysis of the coding of phenomena, the selection of key features of phenomena, and the presentation of professional findings. He looks at the coding, highlighting, and presentation practices of police officers and the effects of their professional discourse on the outcome of the Rodney King trial. For those interested in the embodied aspect of people working in a skilled field and the way in which their practices are socially situated, this would be a good fit.

Mitchell, T. (2002). Can the mosquito speak? In *Rule of experts: Egypt, techno-politics, modernity* (1-27). University of California Press.

This would be a compelling reading if you are interested in reading more about the role of non-human actants, their relations, and the role of power in motivating and shaping the circulation of various global flows. Here, Mitchell is focused on interrogating how various actants, including mosquitoes, chemicals, hydraulic systems, and the like contributed to the public health crisis posed by the malaria epidemic outbreak that occurred in Egypt in the 1940s.

Nail, T. (2016). *Theory of the border*. Oxford University Press.

Nail's book seeks to offer a theory of borders as a way of directing flows, of people, resources, and the like, tracing their use and enforcement in various implementations throughout history. This would be a good read for anyone who wants to see many more examples of how an infrastructural space or object, such as a wall, a prison cell, a fence, or a hospital, operates to control social movement and thus often relies on technologies of control and surveillance that both arise from and comprise its existence.